

Distance Field Driven Reinforcement Learning for Reactive and Robust Collision Avoidance

Taiga Ikeshima (Tokyo Univ. Science), Teresa Vidal-Calleja, Lan Wu, Sheila Sutjipto (Univ. Tech. Sydney), Francisco J Naranjo-Campos (UC3M), Taisuke Kobayashi (NII/SOKENDAI), Taiki Ishigaki, Eiichi Yoshida (Tokyo Univ. Science)

Introduction and Motivation

Conventional Collision Avoidance Methods

- Artificial Potential Field (APF) [1]
- Sampling-based methods (RRT, PRM, etc.)
- Optimization-based methods

Challenges:

Real-time avoidance in unknown & dynamic environments

RL based Methods

Learning collision avoidance policies autonomously

Challenges:

- Accurate obstacle models required
- Sensitive to sensing errors
- Sim2real gap

Objectives

- EDF-based reinforcement learning for robust real-time collision avoidance
- Zero-shot sim-to-real transfer to a real robot

Proposed Method

RL Algorithm

Proximal Policy Optimization (PPO) [2]

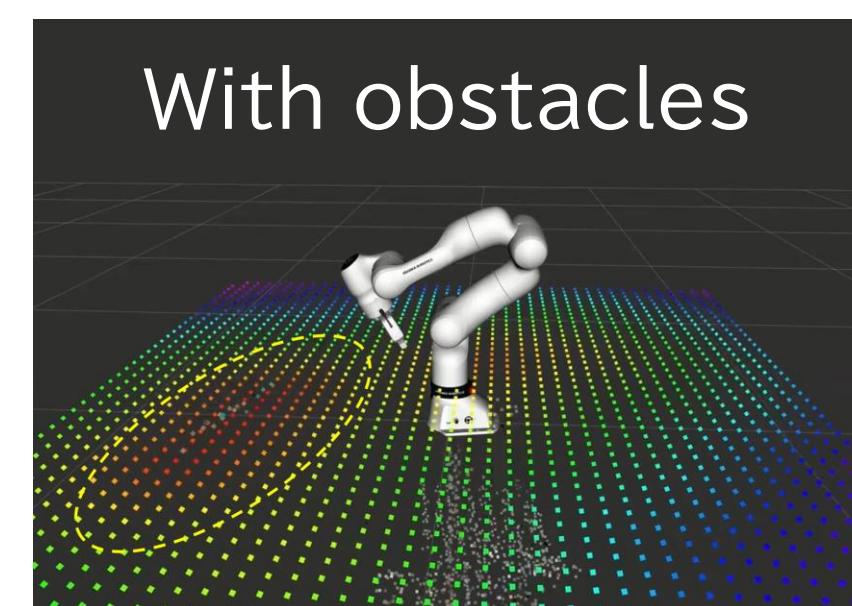
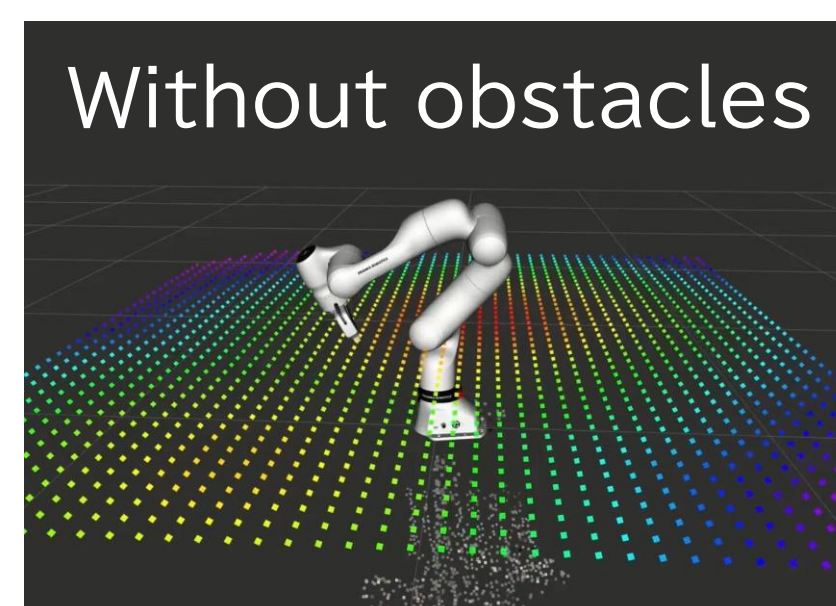
- Using Franka Research 3 model
- Training in MuJoCo simulation

① Euclidean Distance Field (EDF)

- Real-time update from sensor observations
- Continuous spatial representation

Providing:

- Distance to the nearest obstacle
- Gradient (away from obstacles)



XY-plane EDF representation using IDMP [3]

② Observation Space Design

- EE state and goal position
- EDF distance & gradient at EE and arm points (N = 4)

$$\mathbf{o}_t^{\text{FB}} = [\mathbf{p}_{ee}^{\top}, \dot{\mathbf{p}}_{ee}^{\top}, (\nabla d_{ee})^{\top}, d_{ee}, \mathbf{p}_g^{\top}, \{(\nabla d_i)^{\top}\}_{i=1}^N, \{d_i\}_{i=1}^N]^{\top} \in \mathbb{R}^{13+4N}$$

③ Reward Design

Move toward the goal while avoiding obstacles using the EDF gradient

$$R(\mathbf{s}, \mathbf{a}) = -\lambda(\mathbf{s}) d_{ge} + \mu(\cos \sigma - 1)$$

Reference direction:

$$\mathbf{v}_{\text{ref}} = w \frac{1}{d} \nabla d + (1-w)(\mathbf{p}_g - \mathbf{p}_{ee}) + \gamma \hat{\mathbf{o}}, \cos \sigma = \frac{\mathbf{v}_{ee} \cdot \mathbf{v}_{\text{ref}}}{\|\mathbf{v}_{ee}\| \|\mathbf{v}_{\text{ref}}\|}$$

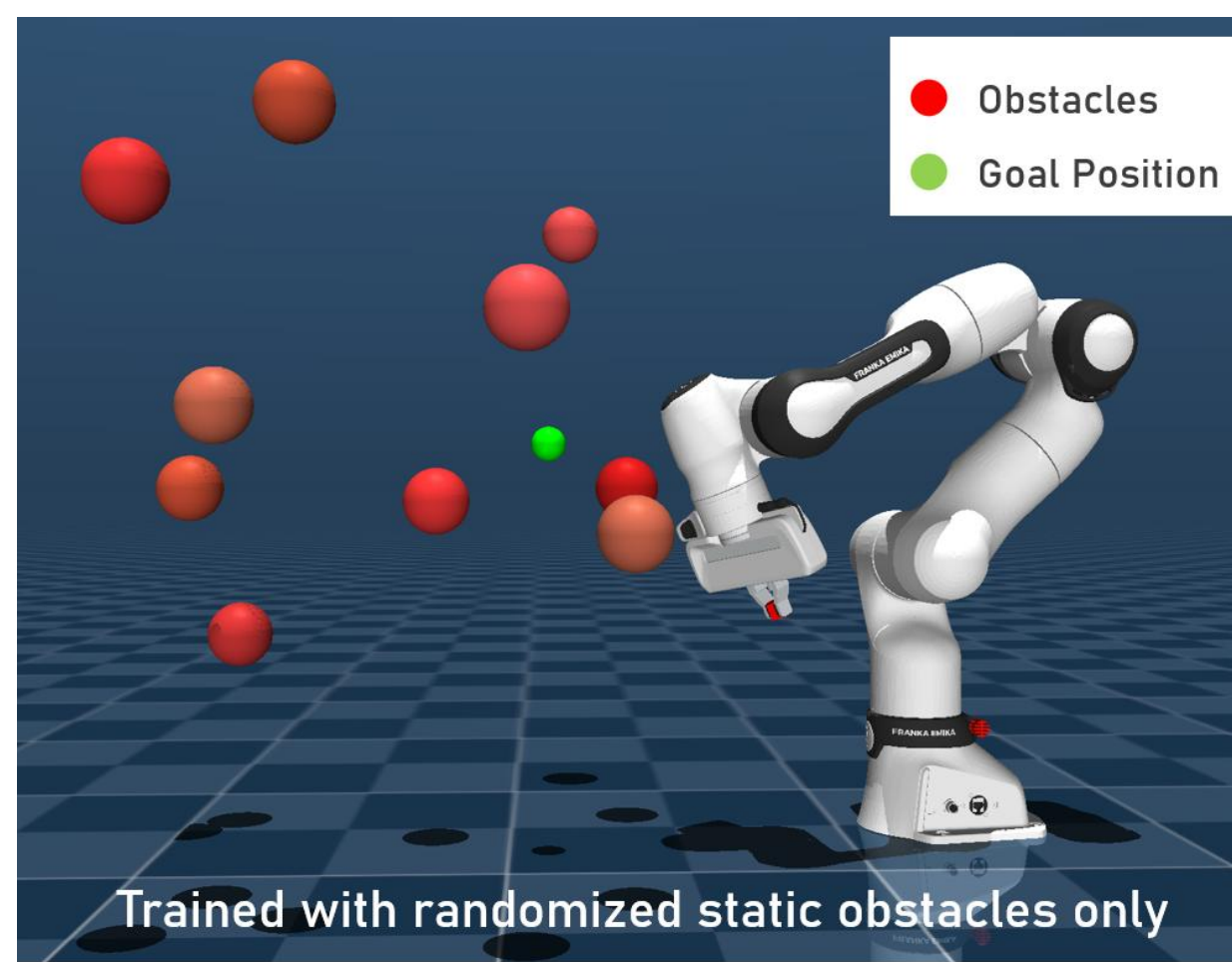
APF-inspired direction: obstacle repulsion + goal attraction [4]



Evaluation

Simulation Evaluation

Task: Reach the goal while avoiding randomly placed obstacles



Trained with randomized static obstacles only

Simulation Results (100 Trials)

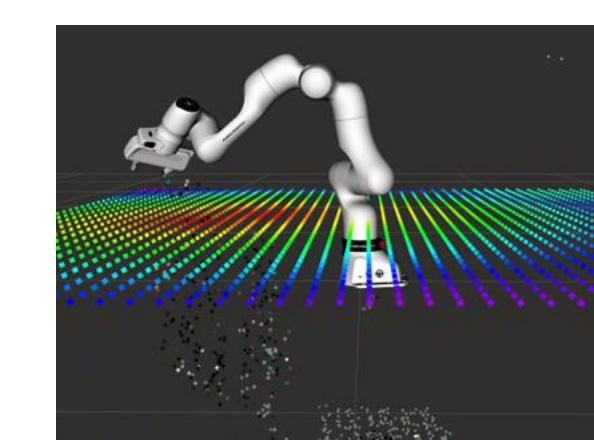
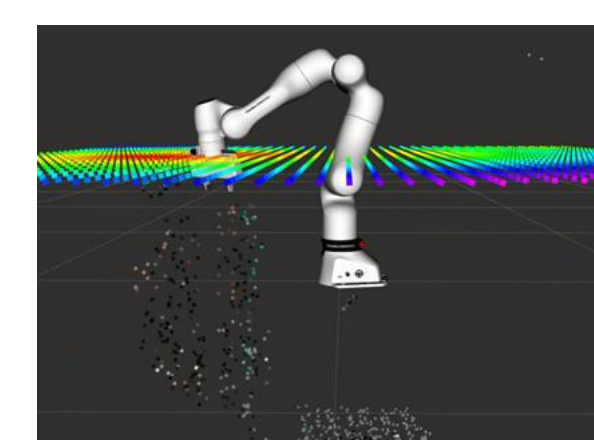
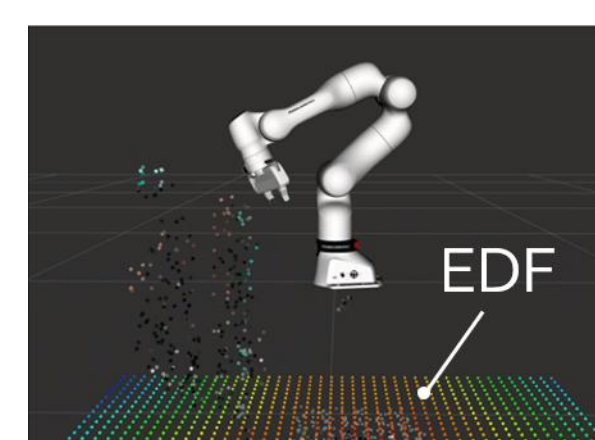
Condition	Reaching goal w/o collision (%)	Reaching goal (%)	Average number of collisions	Reaching time (s) (per episode)
Baseline RL	35.0	51.0	27.9	14.5
RL-APF	64.0	86.0	4.10	9.38
EDF-RL-APF-EE	75.0	96.0	6.79	6.45
EDF-RL-APF-FB	78.0	95.0	3.46	8.52

Training conditions
• PPO (Stable-Baselines3)
• 20M training steps
• 2,000 steps per episode

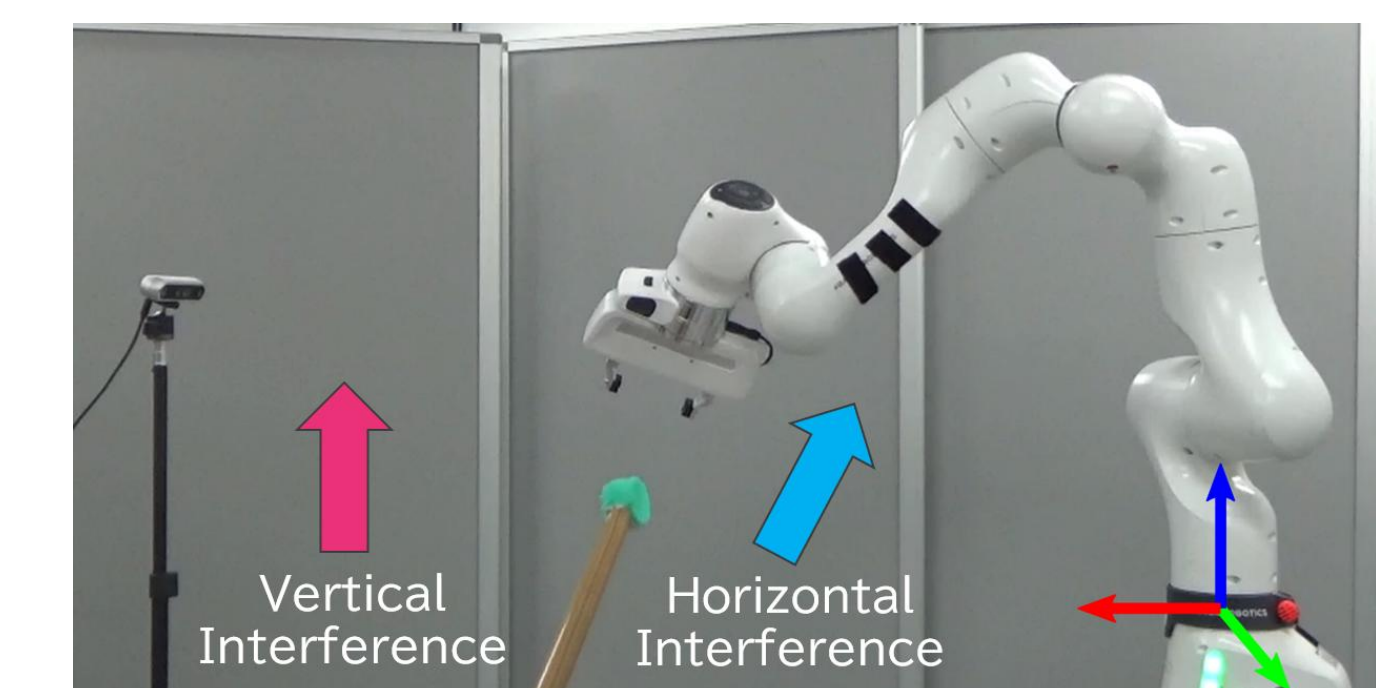
Improved collision avoidance with EDF observations

Real-Robot Evaluation

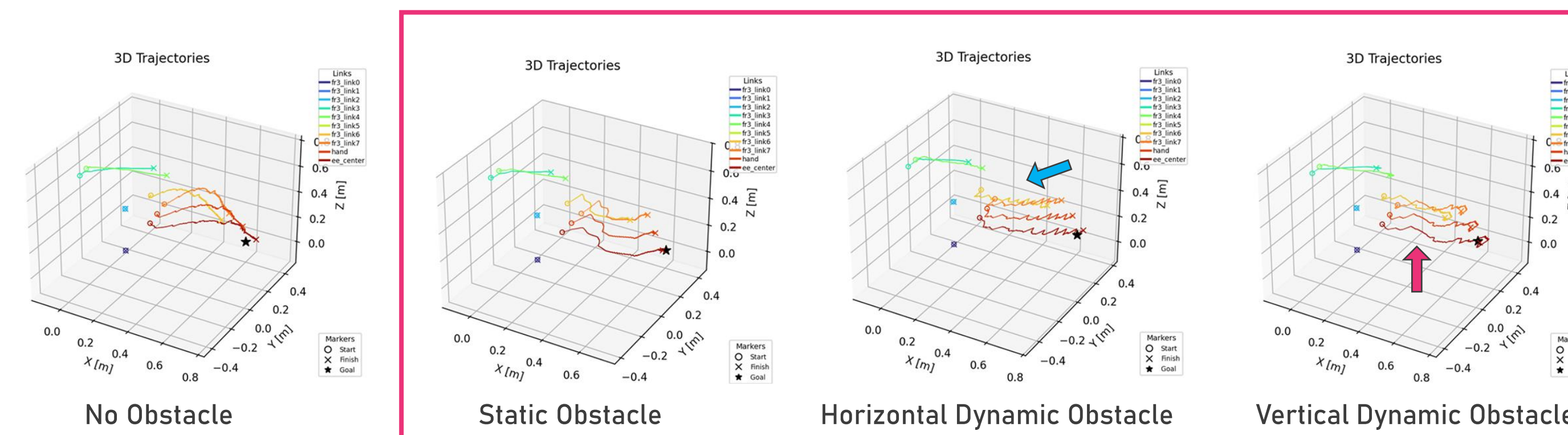
1. Static Obstacles



2. Horizontal & vertical dynamic obstacle interference



Trajectory Comparison across Obstacle Conditions



- Reactive avoidance against static and dynamic obstacles
- Zero-shot sim2real transfer

Conclusion and Future Works

Conclusion

- Integrating EDF geometric information into PPO
- Zero-shot sim2real transfer to a real robot
- Reactive avoidance of unseen/dynamic obstacles
- Validated EDF-based collision avoidance policy

Future Works

- Multi-view RGB-D sensing for occlusion reduction
- Extension of EDF representations to grasping tasks

References

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